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Improving PVT Laboratory Measurements: A Guide to Best Practices

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Abstract

Accurate reservoir fluid characterization relies on minimizing subjective modifications of laboratory measurements using Equations-of-State (EOS). When PVT laboratories encounter unreliable data and lack sufficient oil volumes for re-testing, they often rely on adjustments using EOS. This study proposes a comprehensive review of the PVT standard operating procedures, from sample collection to data reporting, to improve the accuracy and reliability of experimental data, eliminating the need for EOS adjustments and ensuring robust reservoir fluid characterization. Several key enhancements emerged from this proposal, aimed at optimizing laboratory measurements, data collection, verification, and calculations. Notably, these improvements include the implementation of a process-side gauge for accurate pressure measurements, standardizing volume measurement practices, refining equipment setup to achieve "stock tank stage" equilibrium at atmospheric pressure, and streamlining gas sampling processes for floating piston PVT-type cells. Furthermore, supplementary quality control checks are introduced to validate separator pressure and volume measurements at each stage, material balance, and dead oil volume determination. Implementing this proposed procedure will facilitate the straightforward application of raw data to calculate reservoir fluid PVT properties without modifications or EOS corrections. The proposed workflow delivers significant benefits including efficient use of valuable pressurized crude oil samples, and a considerable decrease in PVT experimental repetition rates. Notably, the primary goal of this proposal is to significantly decrease experimental repetition rates. Additionally, by eliminating EOS-based correction, the acquired data truly represents the intrinsic attributes of the reservoir fluids without any theoretical biases. Ultimately, the implementation of these insights enhances decision making in reservoir engineering applications. This study offers effective strategies to overcome equipment limitations. The adoption of the proposed equipment modifications and enhanced procedure enable both researchers and laboratory professionals to utilize best-in-class practices, thereby facilitating the generation of accurate and more reliable PVT laboratory data.

Introduction

Reservoir fluids are naturally occurring complex mixtures of hydrocarbon molecules containing mainly organic compounds (hydrocarbons) and non-hydrocarbons. Hydrocarbons contain carbon and hydrogen molecules, and non-hydrocarbon compounds mainly include nitrogen and acidic gases, which are hydrogen sulphide (H_2S) and carbon dioxide (CO_2). Pressure-Volume-Temperature (PVT) analysis plays a pivotal role in petroleum engineering, offering essential data on the phase behaviour and thermodynamic properties of reservoir fluids under varying pressure and temperature conditions. These properties—such as saturation pressure, oil formation volume factor (B_o), gas-oil ratio (GOR), and fluid density—are integral to key engineering applications, including reserves estimation, surface facility sizing, well completion design, reservoir simulation, and flow assurance (Ahmed, 2016; McCain, 1990; Danesh, 1998).

Accurate PVT characterization of black oil and heavy oil reservoirs begins with the collection of a representative reservoir fluid sample. The accuracy of laboratory measurements depends on how well this sample reflects the original in-situ fluid composition. Given the sensitivity of phase behavior to compositional variation, proper sample acquisition and handling are critical to preserving fluid integrity (Whitson and Brulé, 2000).

The resulting laboratory experimental data inform reservoir management strategies—such as maintaining pressure above saturation in black oil systems—and guide economic decisions by influencing facility design and material selection (John and Flores, 2002; Pedersen et al., 2014). For example, Bo directly impacts original oil-in-place (OOIP) calculations, and GOR impacts the size and design of the processing facilities especially gas-oil separation plant (GOSP). In addition, stock tank API density impacts economic evaluations and revenue projections based on crude oil classification (Riazi, 2005).

Given the significant implications of PVT data across the entire lifecycle of oil and gas projects—from exploration and development to crude oil processing and refining, it is crucial to ensure the accuracy and reliability the generated laboratory measurements. However, these measurements are often subject to inaccuracies arising from equipment limitations, experimental execution, and data interpretation practices. Notably, these errors are utilized by the industry to engage in the practice of subjective tuning of PVT data to fit equation-of-state (EOS) models. However, this can introduce biases, potentially obscuring the true phase behavior of reservoir fluids (Whitson and Brulé, 2000). A key source of measurement uncertainty for floating piston PVT cells stems from the inability to directly monitor pressure within the sample side of high-pressure, high-temperature (HPHT) PVT cells as the pressure gauge is typically on the hydraulic side of the piston to prolong the life of the gauge. This limitation becomes particularly critical at low pressures, where differential pressure between the hydraulic oil and sample side—ranging from approximately 5 to 40 psi—can significantly affect the accuracy of low-pressure and stock tank stages in both separator and differential liberation tests (Firoozabadi, 1999; Whitson and Brulé, 2000).

To address these challenges, emphasis is placed on enhancing test design through targeted equipment modifications, procedural quality checks, and rigorous volumetric calculations. These improvements include the installation of process-side pressure gauges for direct pressure monitoring, refinement of equipment setup to achieve equilibrium at atmospheric pressure, and optimization of gas sampling procedures. Furthermore, additional quality control protocols have been implemented to verify stage-specific volume measurements and refine the accuracy of dead oil volume determination, thereby improving the reliability of material balance calculations.

Comparing experimental data with equation-of-state (EOS) models is a routine practice in PVT laboratories. However, this process often involves adjusting or ‘tuning’ specific data points to achieve better alignment with the EOS model, which can trigger ripple effects that require further modifications to other parameters. This tuning process is inherently subjective and is typically guided by the judgment of the data quality engineer conducting the review, assuming the deviation from the model is coming from accepted experimental errors of the PVT system (Danesh, 1998). Although these adjustments aim to improve the model fit, they may inadvertently obscure true deviations in fluid behavior, particularly when these deviations are attributed to the inherent properties of the fluid rather than measurement uncertainty. The engineer's ability to accurately identify the source of deviations depends largely on their expertise, access to historical data for benchmarking, and the capability of the EOS model to effectively represent the fluid's behavior. Consequently, the final dataset, when used by reservoir engineers, may contain undetected errors that could influence decision-making. Moreover, the adjusted data, once archived, becomes problematic for future studies, as it serves as a benchmark for comparison with new data, potentially perpetuating long-term errors in reservoir analysis and modeling (Ahmed, 2016).

Overview of PVT Testing

PVT analyses form a fundamental part of reservoir characterization, with the primary objective of quantifying key fluid properties including bubble-point pressure—the pressure at which the first gas bubble emerges from solution, formation volume factor (Bo)—the ratio of the volume of a single-phase fluid mixture at a given pressure and temperature, typically reservoir conditions, to the volume of the same fluid when brought to surface conditions of 1 atm and 60°F, and solution gas-oil ratio (GOR) —the amount of gas dissolved in the oil. These measurements are obtained through a series of controlled laboratory experiments conducted in high-pressure, high-temperature (HPHT) PVT cells (McCain, 1990; Firoozabadi, 1999). A standard black oil PVT study includes the constant composition expansion (CCE), differential liberation (DL), and multistage separator (ST) tests. The CCE test is typically conducted first and is non-destructive, whereas the DL and ST tests are destructive that are generally conducted subsequent to the CCE test.

The CCE test is conducted to determine the bubble-point pressure and volumetric behavior of reservoir fluids, typically at reservoir temperature as shown in Figure 1. In this experiment, the fluid undergoes a stepwise pressure reduction from an initial value above the reservoir pressure, while the temperature is held constant to simulate isothermal expansion. Key outputs from the CCE test include the bubble-point pressure, liquid compressibility, and relative volume. These parameters are fundamental for characterizing the fluid's phase behavior and provide a baseline for subsequent phase behavior tests (Danesh, 1998; Whitson and Brulé, 2000).

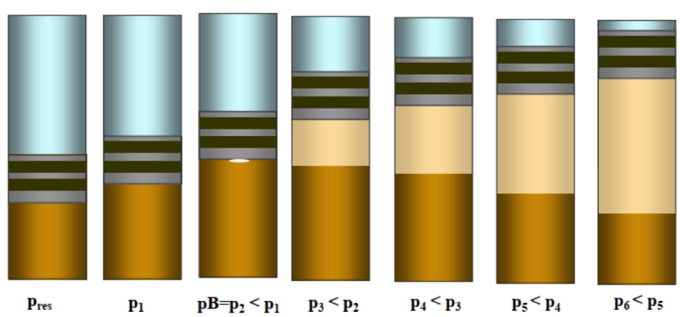

Figure 1—Visual of the Constant Composition Expansion (CCE) Test

Following the CCE, either a multistage separator or differential liberation test is performed. The separator test simulates the surface separation process, typically as it occurs in a GOSP as shown in Figure 2. Performed in one or more stages, at predefined pressure and temperature for each stage, the separator test provides stage-specific and overall measurements of GOR, B_o, and gas composition, along with stock tank API gravity. The B_o obtained from this test enables the conversion between reservoir and surface volumes, supporting accurate reserves estimation and production allocation (Ahmed, 2016; Pedersen et al., 2014). Gas compositions at each separation stage are analysed using gas chromatography, allowing detailed quantification of both non-hydrocarbon components (e.g., N_2, CO_2, H_2S) and hydrocarbons (C_1, C_2, C_3, ……, C_{n+}). The density of the stabilized oil at stock tank conditions is measured after the final separation stage, reported as API gravity, is an indicator of crude oil quality.

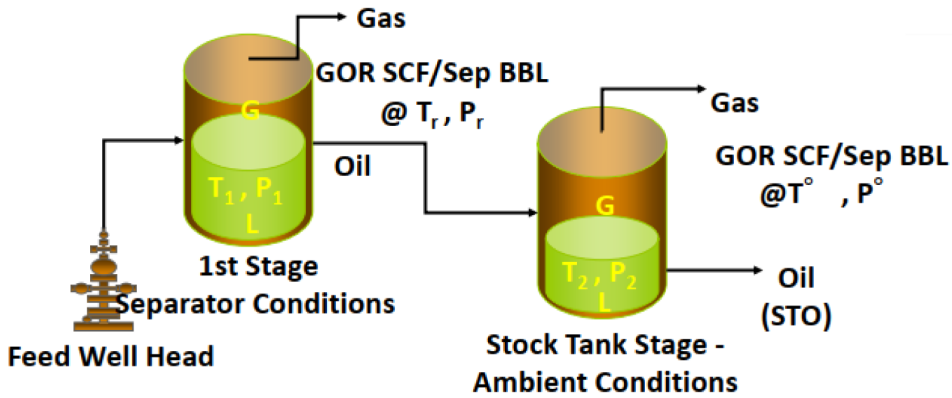

Figure 2—Visual of the Multistage Separator Test

The differential liberation test, while yielding similar output parameters, differs from the separator test in that all stages are conducted at a constant temperature, typically representative of reservoir conditions. This test simulates reservoir depletion under natural production without external pressure support, offering insight into the behavior of the reservoir fluid as pressure declines. Although the same key parameters—stage-specific and overall GOR, Bo, and gas composition, along with stock tank API gravity—are reported, their values differ from those obtained in the separator test due to the influence of temperature on phase behavior (Firoozabadi, 1999).

Together, these tests form the core of black oil PVT studies, providing a comprehensive understanding of reservoir fluid properties under both surface and subsurface conditions. The reliability of the resulting data hinges on rigorous test execution and adherence to best practices in laboratory procedures.

PVT Equipment Limitations

Accurate PVT measurements depend heavily on both the precision and reliability of the laboratory equipment. Below, we delineate the major equipment limitations observed in commonly used PVT systems, focusing on how each limitation impacts thermodynamic parameters and the overall integrity of PVT studies. When conducting PVT experiments, the temperature is usually fixed at either reservoir or stage-specific temperature, and the key measured properties are pressure and volume; obtained by adjusting the cell's volume or pressure respectively. These measurements are then used for subsequent calculations to obtain key parameters such as bubble point pressure, formation volume factor and compressibility (Ahmed, 2016; McCain, 1990).

There are various types of PVT cells available on the market, all operating based on similar principles. These cells are pressure-rated vessels designed to contain fluid samples, with mechanisms to manipulate pressure, temperature, and cell volume to study the impact of these variables on fluid behavior. Typically, temperature is controlled using heating bath ovens, pressure is regulated via pumps, and volume is adjusted using an internal piston. The cells are also equipped with relevant sensors such as pressure gauges, temperature probes, and mechanisms to determine the location of the piston for determining the sample's volume (Danesh, 1998; Whitson and Brulé, 2000).

A key distinction among different PVT cells lies in the design of the piston—specifically whether it is a "driven" or "floating" piston. A driven piston is actively controlled by a motor or actuator, enabling precise volume adjustments and measurement. This control is particularly beneficial for dynamic and automated tests, such as swelling tests, separator simulations, and volumetric measurements. However, driven piston systems are mechanically complex, more expensive, and require regular calibration and maintenance due to their active components. In contrast, a floating piston operates passively, relying on differential pressure between the reservoir fluid and a pressurizing medium—typically hydraulic oil. While this design is simple, cost-effective, and low maintenance, it provides limited control over piston movement and can be prone to differential sticking. This can result in pressure inaccuracies during in-situ stage-specific conditions or may cause hydraulic fluid leakage into the reservoir sample if not properly maintained. Despite these limitations, floating pistons are still widely used in PVT laboratories due to their simplicity and cost-efficiency.

Piston-Drag and Pressure Disequilibrium

There are inherent limitations in PVT cells that can have significant implications on the accuracy of test results. *Pressure disequilibrium* due to "Piston-Drag" is one of the primary challenges associated with the use of floating pistons PVT type cells. The piston separates between the pressurizing fluid side (hydraulic fluid) and the reservoir fluid sample within the cell. A seal, usually an elastomer O-ring, is used to ensure that the sample is preserved, and does not interact with the hydraulic fluid. "Piston-Drag" occurs due to frictional forces between the piston and the cell wall, preventing the piston from moving freely. As the fluid inside the cell expands or contracts, differential pressure must be applied to move the piston. However, this drag-force can result in inaccurate volume and pressure readings on the sample side, particularly at low pressures. The displacement of the piston in the cell directly correlates with the volume of the sample fluid. The position of the piston is continuously monitored using a position transducer or cathetometer. The sensors are calibrated to convert the position of the piston into a volume reading. In ideal conditions, negligible differential pressure would allow for piston movement. In practice, a differential pressure of 5–10 psi is often required to overcome piston drag, which increases over time as the seal material swells due to exposure to corrosive fluids, H_2S or CO_2. Over the course of a multi-day study, this can increase the differential pressure needed to move the piston to as high as 30–40 psi. This discrepancy in pressure can lead to significant errors, as the actual pressure inside the cell deviates from the pressure gauge reading that is connected to the hydraulic fluid side. A summary of the ideal and actual depressurization of a saturated fluid during a stage test is shown in Figure 3. Many labs using floating PVT cells capture their study pressure from the hydraulic pressure as this hydraulic fluid is usually not corrosive or straining the gauge internals. It requires less maintenance for the lab and less risk of calibration errors of the gauge. Therefore, the stage-specific intended study conditions are not achieved, especially at low pressures. The consequence is that erroneous data may be confidently reported, ultimately leading to inaccurate stage-specific gas composition, GOR, and B_o (Dodson et al., 1954; Ahmed, 2016).

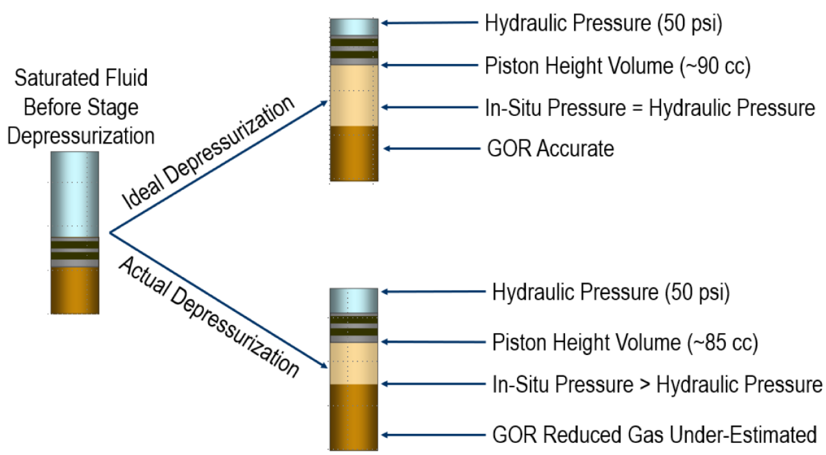

Figure 3—Piston Drag Effect on Pressure-Volume Equilibrium and Measurements

Gas Evolution and Stock Tank Stage Equilibrium

One of the primary challenges in PVT analysis is achieving true equilibrium at the final stage of the differential and separator tests, commonly referred to as the stock tank stage. Ideally, by this point, all dissolved gases should have fully evolved, leaving only the residual stock tank oil within the PVT cell. This stage is critical for quantifying the total evolved gas and capturing representative samples for compositional analysis. However, accurately determining the *volume and gas composition of the stock tank stage* presents significant technical challenges. To sample the evolved gas, the PVT cell must be opened, and the piston advanced to displace the remaining gas into a collection vessel, such as a pycnometer or gasometer. In practice, this process is complicated by piston drag—an inherent limitation in a floating piston cell. Although the hydraulic side of the piston is set to atmospheric pressure, the internal reservoir fluid often lacks sufficient force to overcome the static friction imposed by the piston seal. As a result, the piston may fail to move as intended, preventing the system from reaching true atmospheric equilibrium at the stock tank conditions. Furthermore, when gas evolution exceeds the volumetric capacity of the PVT cell, the excess gas is usually diverted to an external gasometer. After stabilization at atmospheric pressure, the gas volume is measured for use in gas-oil ratio (GOR) and material balance calculations. A portion of this gas is then directed to a gas chromatograph (GC) for compositional analysis. However, the sampling process may introduce additional sources of error. The transfer lines between the cell and the gasometer often contain substantial dead volumes. To push the remaining gas through these lines, additional differential pressure must be applied, which can inadvertently re-dissolve portions of the evolved gas back into the stock tank oil—thereby disrupting the equilibrium state. Additionally, if the transfer lines are not properly heated, condensation of heavier hydrocarbons may occur, leading to unrepresentative sampling. These issues often result in gas samples that are compositionally skewed—typically enriched in lighter components such as methane (C_1) and depleted in intermediate fractions like ethane to pentane (C_2–C_5). In such case, comparisons with EOS model predictions reveal significant deviations. In response, laboratories may repeat the analysis to validate the data; however, because the same procedures and equipment are typically employed, the error is systematic and tends to persist across repeated runs. Consequently, the reproduced data may continue to misrepresent the true fluid behaviour, thereby undermining the integrity and reliability of PVT studies used for reservoir characterization (Danesh, 1998; Pedersen et al., 2014).

Dead Oil Volume Recovery

Finally, the recovery of dead oil volume remains a critical concern in PVT studies. During the stock tank stage, some residual oil inevitably remains trapped in the cell's dead volumes, as well as in the sampling loop and other components of the system. This residual oil is not captured during the sampling process, resulting in a material loss that impacts material balance calculations. Furthermore, during the separation stages, as gas is fully removed to leave only the saturated fluid, some fluid is introduced into the gas sampling coil (i.e., the dead volume in the gas line). To maintain the integrity of the sampling process, the oil in the sampling loop must be purged after each separation stage to ensure that only clean gas is collected. However, this purging step further diminishes the accuracy of material balance calculations. The unaccounted-for oil volumes in these dead spaces can lead to significant errors, particularly in the determination of formation volume factor (B_o) and other key PVT parameters (McCain, 1990; Riazi, 2005).

In summary, while PVT equipment is essential for understanding reservoir fluid behaviour, there are several limitations inherent to both the design of the cells and their operation that can significantly impact the accuracy of the data. Addressing challenges such as piston-drag, achieving equilibrium at the stock tank stage, ensuring accurate volumetric measurements, and recovering dead oil volumes is crucial to improving the reliability of PVT studies.

Mitigating Equipment Limitations

Given the inherent limitations of PVT laboratory equipment, especially floating piston cells, it is essential to implement targeted mitigation steps to enhance data accuracy and reliability. These steps focus on overcoming challenges related to pressure measurements on the sample side, gas sampling, volumetric accuracy, and material balance. Here, we outline key best practices and how they directly address these challenges to ensure more precise and repeatable experimental PVT results.

Implementation of In-Situ Pressure Gauge

One of the critical factors in obtaining accurate PVT measurements is ensuring that the pressure at each stage of the experiment is accurately achieved. In typical PVT testing, pressure discrepancies between the hydraulic side of the cell and the sample side can lead to significant errors in subsequent measurements, such as gas composition and residual oil volumes.

To address this, an in-situ process-side pressure gauge was installed on the sample side of the PVT cell. This gauge allowed laboratory personnel to achieve equilibrium at the desired stage pressure. Unlike hydraulic-side gauges, which measure the pressure on the hydraulic fluid side, in-situ gauges measure the true internal pressure of the reservoir fluid sample, ensuring that the study is conducted at the correct pressure condition. This practice eliminates potential errors introduced by piston drag in floating piston cells, particularly in low-pressure stages where small pressure discrepancies can drastically affect the accuracy of measurements.

The adoption of an in-situ gauge does, however, introduce new challenges. These gauges require more frequent calibration and can be exposed to aggressive fluids, necessitating rigorous maintenance and cleaning protocols. To address these challenges, regular quality checks and calibrations of gauges are implemented, as well as detailed HSE assessments to ensure that the metallurgy is compatible with the fluids being studied. Despite the increased operational costs, the improved data quality justifies the investment, as it reduces the risk of inaccurate data and the need for repeat studies.

Enhanced Volumetric Quality Control

Volumetric calculations during separator or differential liberation stages traditionally rely on measurements from the PVT cell, where piston displacement is used to estimate the fluid volume. A position transducer continuously monitors the piston's position, converting it into a volume reading. However, issues such as piston-drag can introduce errors in these measurements. To improve accuracy, a modified procedure incorporating a Constant Composition Expansion (CCE) step—referred to as mini-CCE—has been introduced after each separation stage.

In this approach, after the gas phase is expelled, the pressure in the cell is increased, followed by controlled decompression, which simulates the CCE process. During the mini-CCE, pressure is gradually reduced, and volumetric readings are recorded at multiple pressure points. This enables an independent and more accurate determination of the saturation pressure. If all gas has evolved at the stage-specific pressure, the resulting saturation pressure from the mini-CCE should match the stage pressure, confirming the equilibrium conditions have been achieved.

The mini-CCE also validates the single-phase volume at the stage conditions, improving the precision of volumetric measurements, particularly when combined with in-situ pressure gauge data. Furthermore, comparing the stage pressure with the saturation pressure from the mini-CCE ensures that true equilibrium has been achieved, thus enhancing the reliability of the volumetric data and providing a robust quality control mechanism in PVT testing.

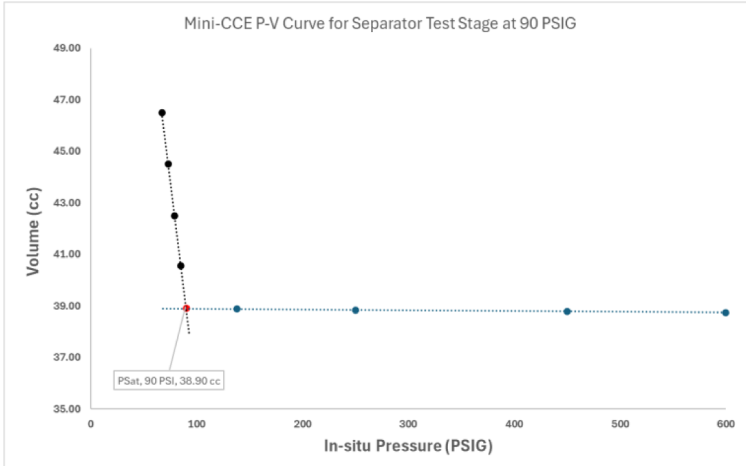

Figure 4—An Example of a Mini-CCE Pressure vs. Volume Curve for a Typical Stage (Separator or Differential-Liberation) Test

Design of Stock Tank Gas Sampling Protocol

Accurately capturing the gas composition at the stock stank stage is a critical step in PVT analysis, yet it is often prone to errors, particularly in terms of residual gas that remains trapped in the cell. To overcome this, a calibrated external expansion volume was incorporated above the PVT cell, allowing for the gas cap to be captured and equilibrated without the need to forcibly displace gas through pressurization, which can distort the gas composition. This modification involves opening the cell to the additional expansion volume after the piston has stopped moving, allowing the gas cap to freely travel into the expansion volume. The cell is then allowed to re-equilibrate, ensuring that the gas in the external volume is at equilibrium with the reservoir fluid. This prevents the re-saturation or compositional distortion that typically occurs when gas is pushed under pressure.

Once equilibrium is reached, the gas is directly injected into a gas chromatograph for compositional analysis, bypassing issues related to dead volumes and potential compositional errors. Given that the external expansion volume is located within the oven, the gas is heated and maintains its homogeneity, thereby preventing any condensation of the hydrocarbon components. This method has proven effective in correcting discrepancies observed in C_2–C_5 components, which are often the source of significant deviations from EOS model predictions in the stock tank stage as shown in Figure 5. The figure shows that utilizing the gas cap expansion (blue line) leads to closer compositional match with the EOS (dashed line) in comparison to the traditional piston push off (red line).

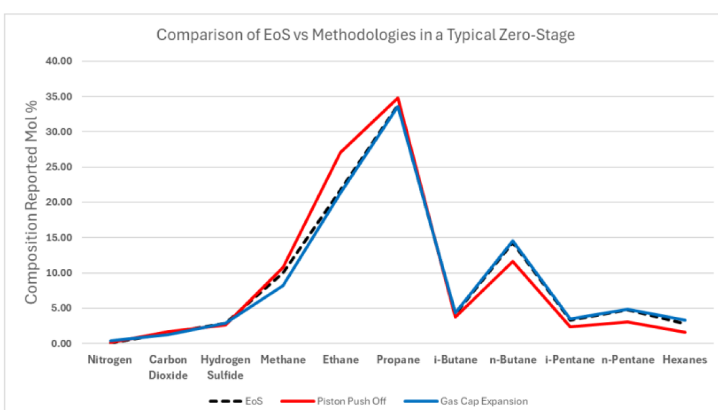

Figure 5—Zero Stage Compositions from a Typical Black Oil Separator Test using Expansion (blue) vs Traditional (Red) Methods

Correction of Material Balance and Stock Tank Conditions

A common challenge in PVT studies is the loss of material due to residual fluid trapped in the dead volumes of the PVT cell, the sampling loop, or other internal components. This loss of fluid can lead to significant errors in material balance calculations, particularly for key parameters like formation volume factor (Bo) and gas-oil ratio (GOR).

To address this, glass traps were installed to capture any expelled oil or gas during the purging process. The captured fluids are measured gravimetrically, and their volumes are reintegrated into the material balance, ensuring that the total fluid volume is accurately accounted for. In cases where the PVT cell cannot achieve standard conditions, post-test density measurements using calibrated densitometers are conducted to correct the dead oil volume to standard conditions. As a result, the material balance target of less than 3% is achieved, which is within acceptable experimental error of less than 5%.

Procedural Transparency and Raw Data Integrity

The implementation of enhanced practices is going to significantly improve transparency and standardize calculation methodologies across laboratories and clients. By providing raw, unprocessed data, laboratories enable clients to independently audit the process, ensuring no subjective tuning of experimental data, unless explicitly requested. This transparency allows clients to assess data integrity and gain confidence in the results, even when natural deviations from EOS models are observed. As a result, a notable decline in experimental repeat analysis requests is expected, as clients are better equipped to understand the reasons for any observed deviations.

In addition to enhancing transparency, these proposed practices address inherent equipment limitations and refine procedural protocols, leading to more accurate and reliable fluid characterization. Although these enhancements will require additional resources, they represent a critical investment in ensuring that PVT data meets the rigorous standards required for reservoir modelling and economic evaluation.

The education and training of operators play a critical role in effectively utilizing raw PVT data without subjective tuning. While the adoption of the proposed best practices minimizes experimental errors, it is essential to recognize that some degree of measurement uncertainty is inherent in PVT studies. Operators must therefore possess a comprehensive understanding of the PVT process to interpret data accurately and acknowledge the limitations of experimental measurements. This understanding foster acceptance of the natural variability that may arise during testing. To achieve this, a collaborative approach between service companies and clients is essential. Service laboratories should commit to continuously improving measurement accuracy and procedural transparency, while clients must recognize that raw PVT data may not perfectly align with EOS model predictions. This mutual understanding ensures that the data is interpreted within the appropriate context, leading to more reliable reservoir characterization and decision-making.

Conclusion

This study underscores the critical importance of obtaining accurate and transparent PVT measurements. This paper presents a systematic approach to address PVT equipment limitations in order to improve data quality, reduce subjective tuning, and enhance the overall reliability of reservoir fluid characterization. The implementation of the proposed best practices—such as in-situ pressure gauges, mini-CCE volumetric control, and rigorous material balance protocols—is going to lead to significant improvements in laboratory efficiency and data integrity. The enhanced quality check processes is going to reduce repetition rates, improve turnaround times (TAT), and lower the consumption of valuable samples.

These practices will improve the reproducibility and accuracy of fluid properties. This leads to increased operational efficiency, which translate to substantial cost savings and faster decision-making. Furthermore, prioritizing transparency through raw data reporting allows clients to independently validate results, leading to increased trust and auditability. These advancements in PVT measurements directly contribute to better reservoir modelling and production strategies, enabling more efficient hydrocarbon extraction. The elimination of subjective EOS adjustments ensures data integrity, minimizing biases that could otherwise lead to suboptimal decisions.

In conclusion, the resolution of equipment-related limitations in PVT testing offers substantial technical and economic value. The methods described here not only reinforce the critical role of high-quality PVT data in reservoir and production engineering but also provide a roadmap for laboratories and operators seeking to optimize data quality, reduce rework, and improve project outcomes.

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